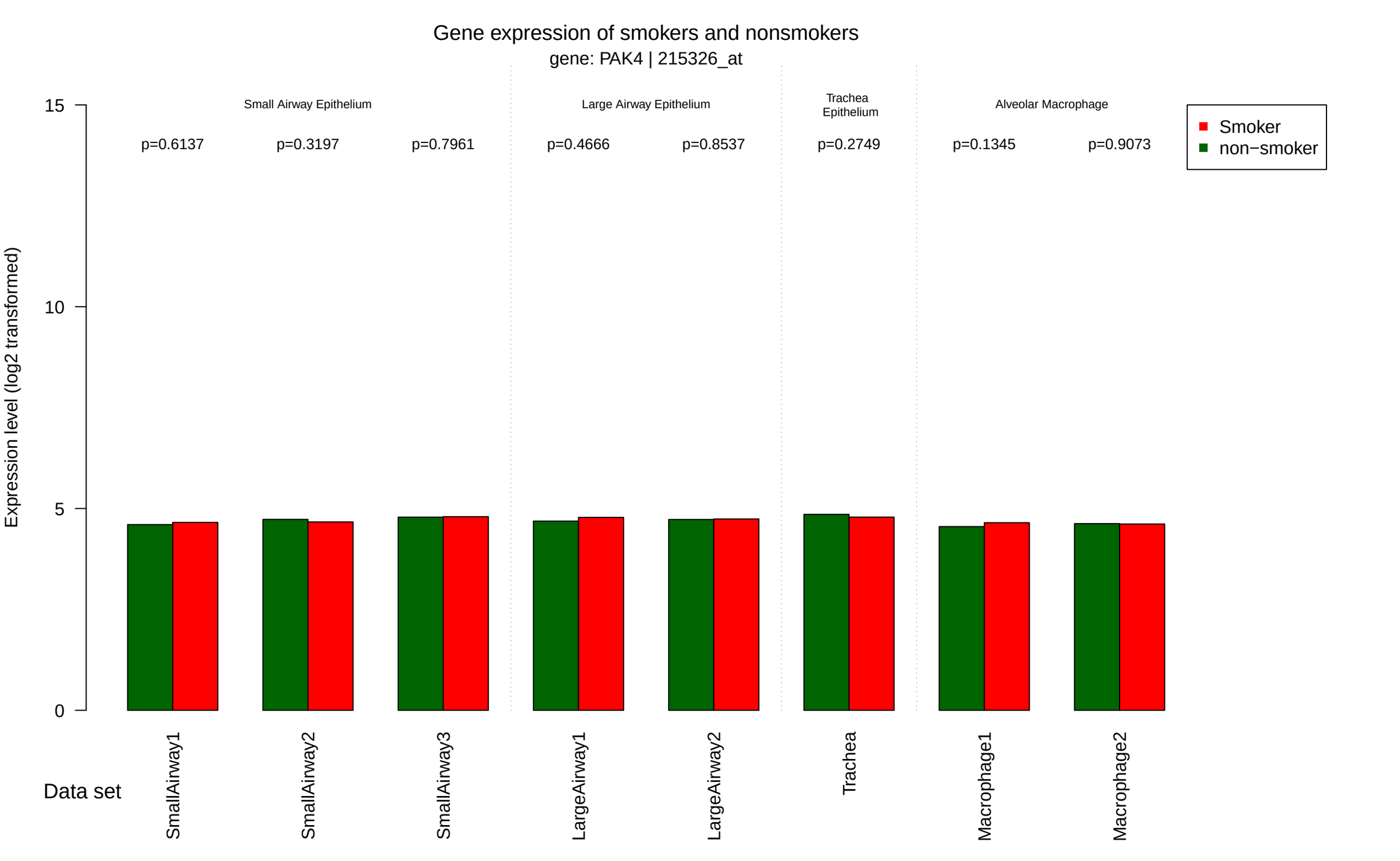
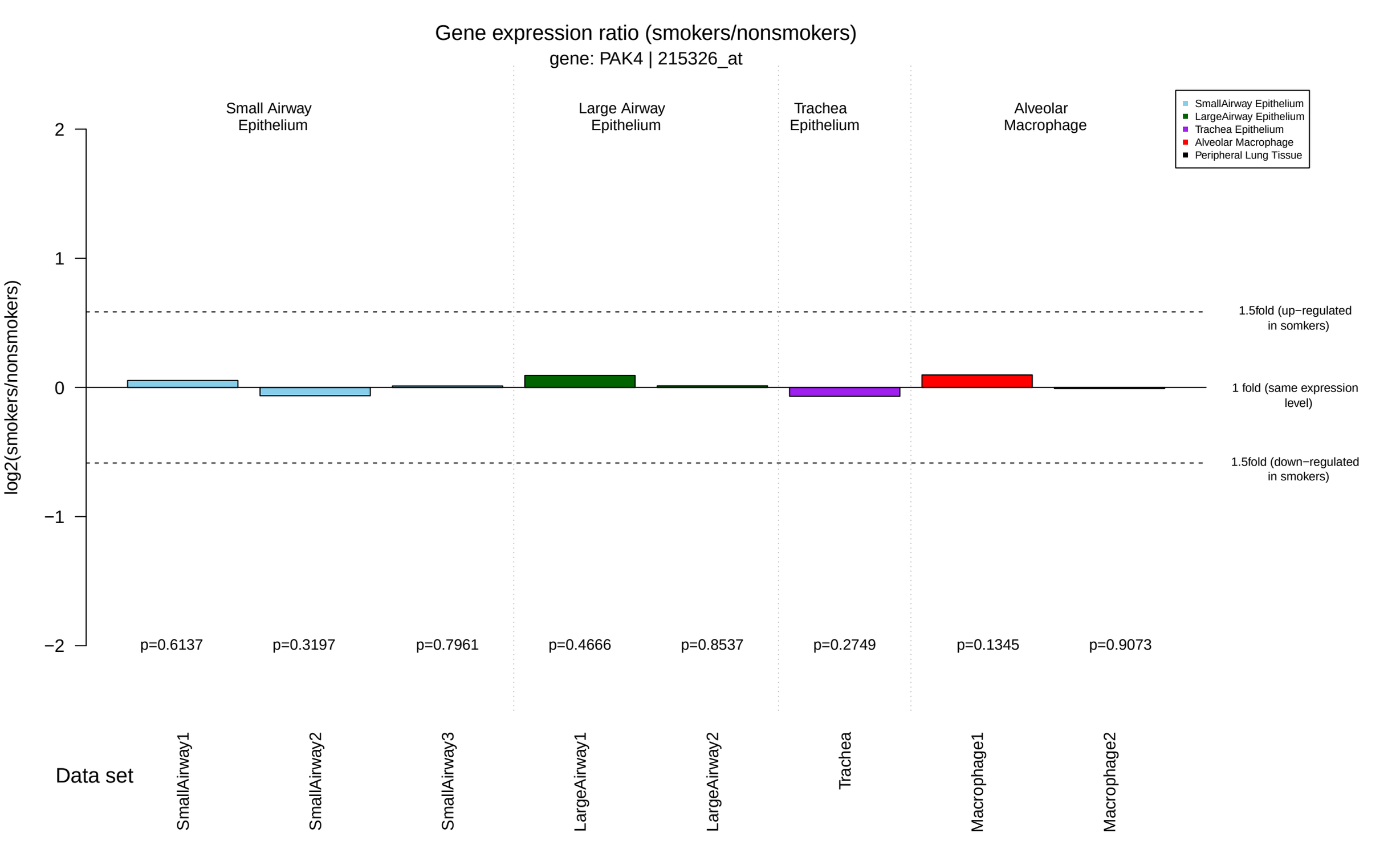
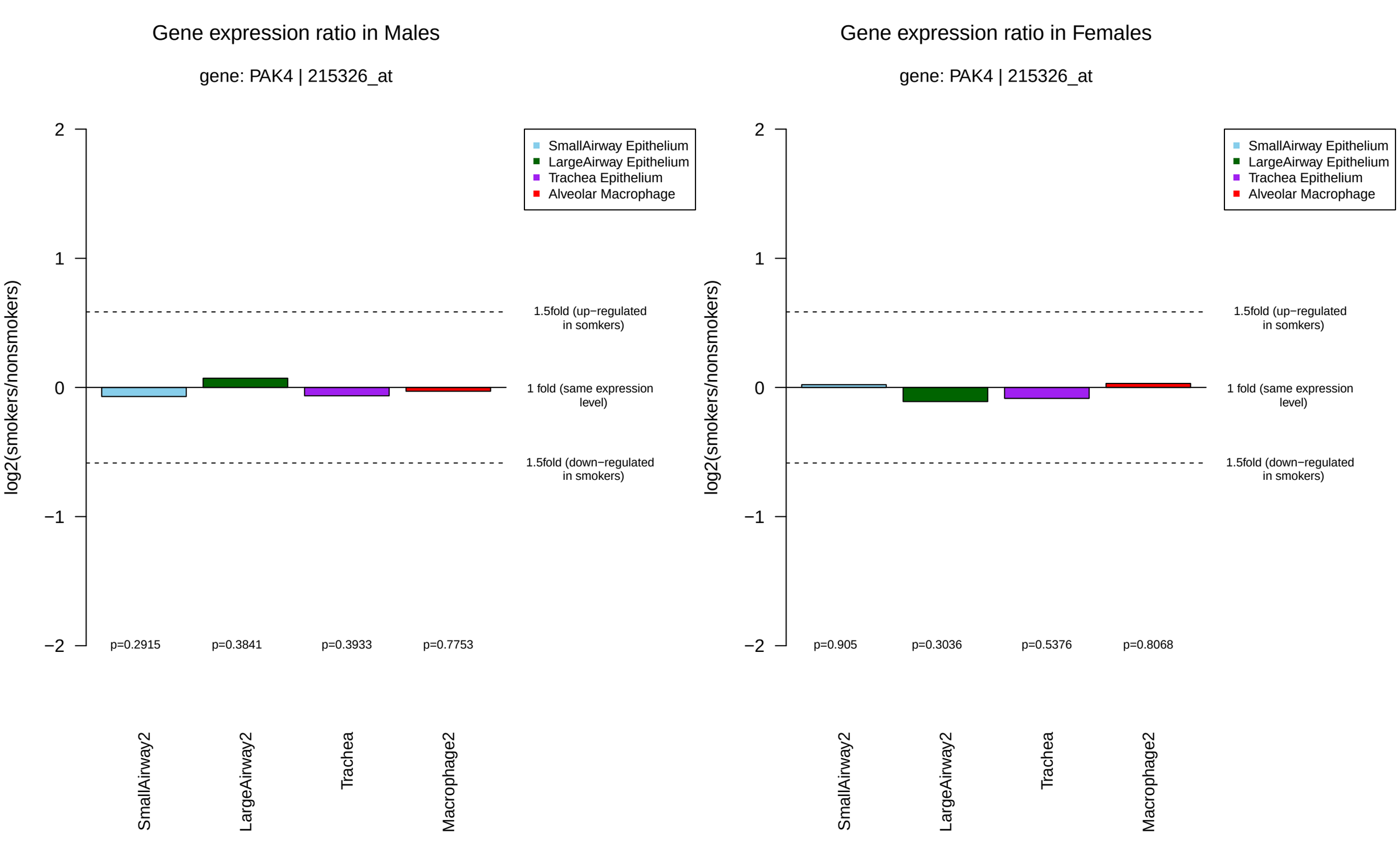
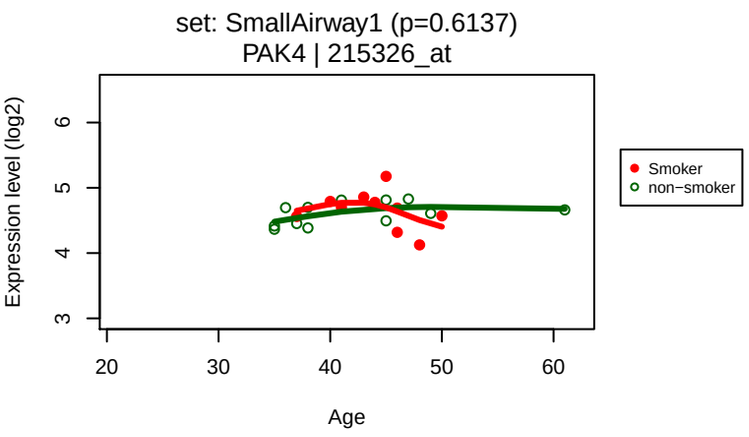
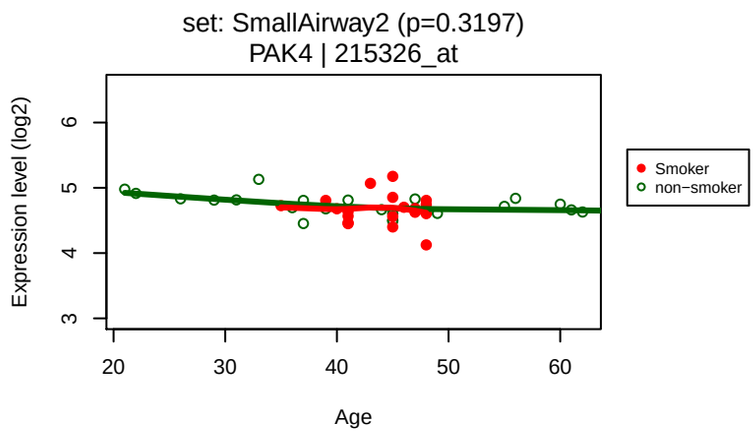
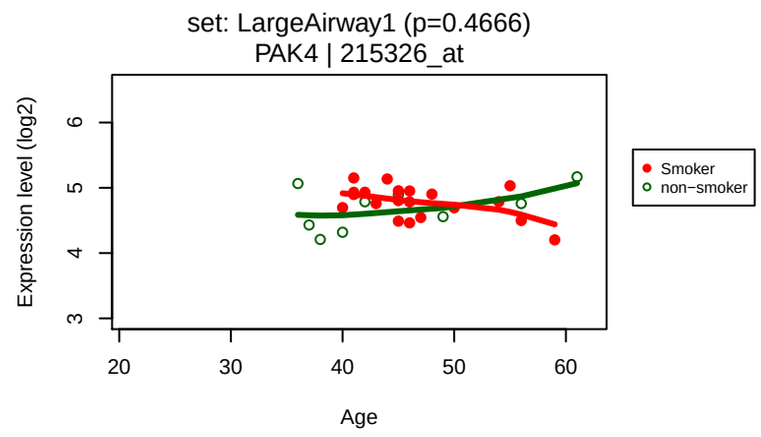

Gene expression of smokers and nonsmokers
gene: PAK4 | 215326_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6137
p=0.3197
p=0.7961
p=0.4666
p=0.8537
p=0.2749
p=0.1345
p=0.9073
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

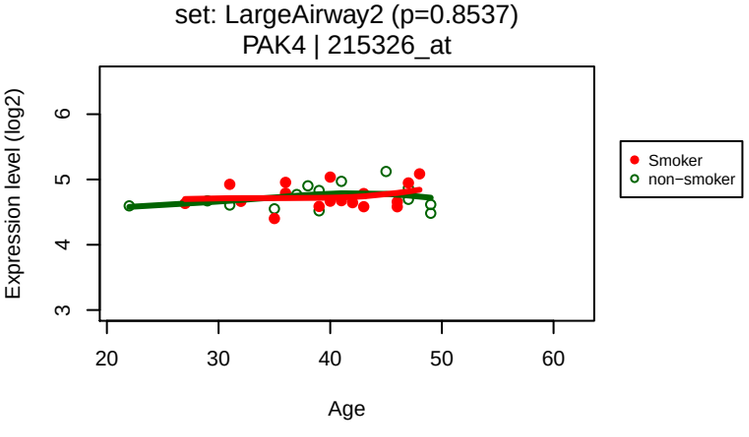

Gene expression ratio (smokers/nonsmokers)
gene: PAK4 | 215326_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6137
p=0.3197
p=0.7961
p=0.4666
p=0.8537
p=0.2749
p=0.1345
p=0.9073
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

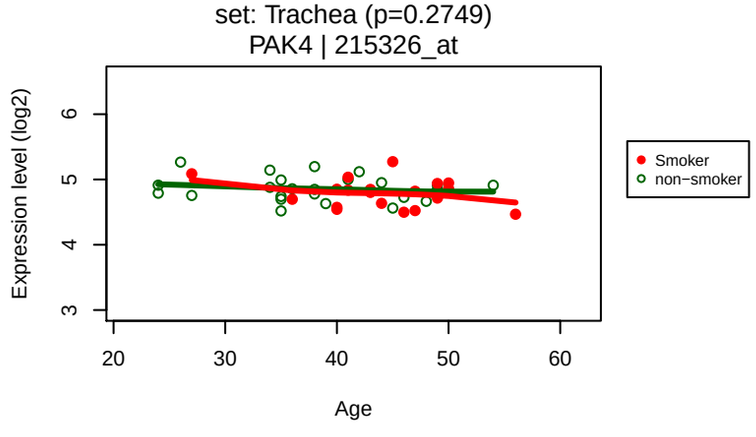

Gene expression ratio in Males
gene: PAK4 | 215326_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2915
p=0.3841
p=0.3933
p=0.7753
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK4 | 215326_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.905
p=0.3036
p=0.5376
p=0.8068
SmallAirway2
LargeAirway2
Trachea
Macrophage2

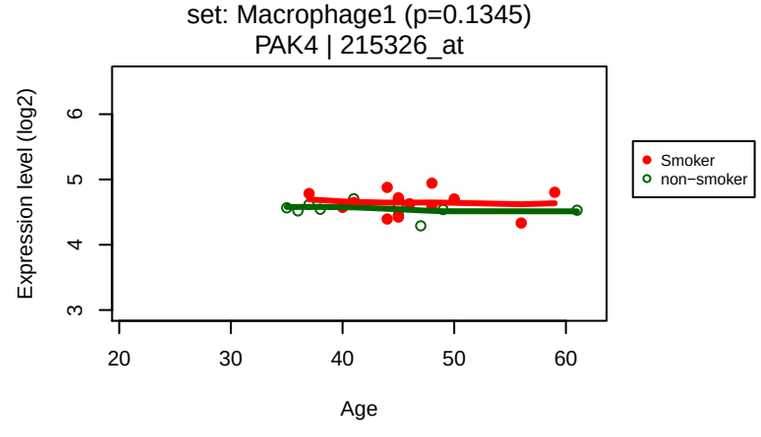

set: SmallAirway1 (p=0.6137)
PAK4 | 215326_at
Expression level (log2)
Age
Smoker
non−smoker

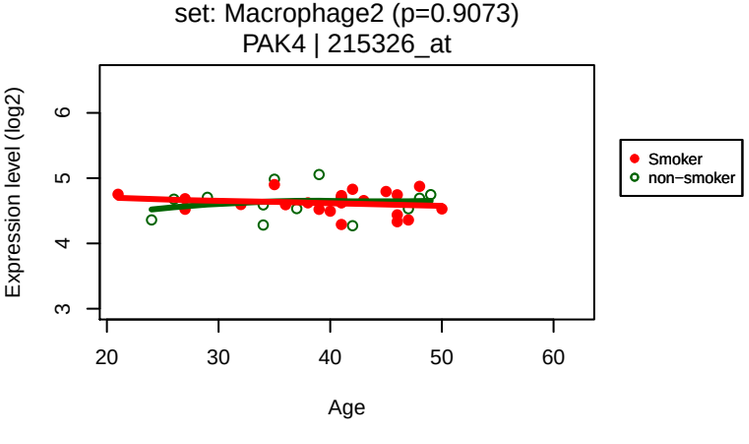

set: SmallAirway2 (p=0.3197)
PAK4 | 215326_at
Expression level (log2)
Age
Smoker
non−smoker

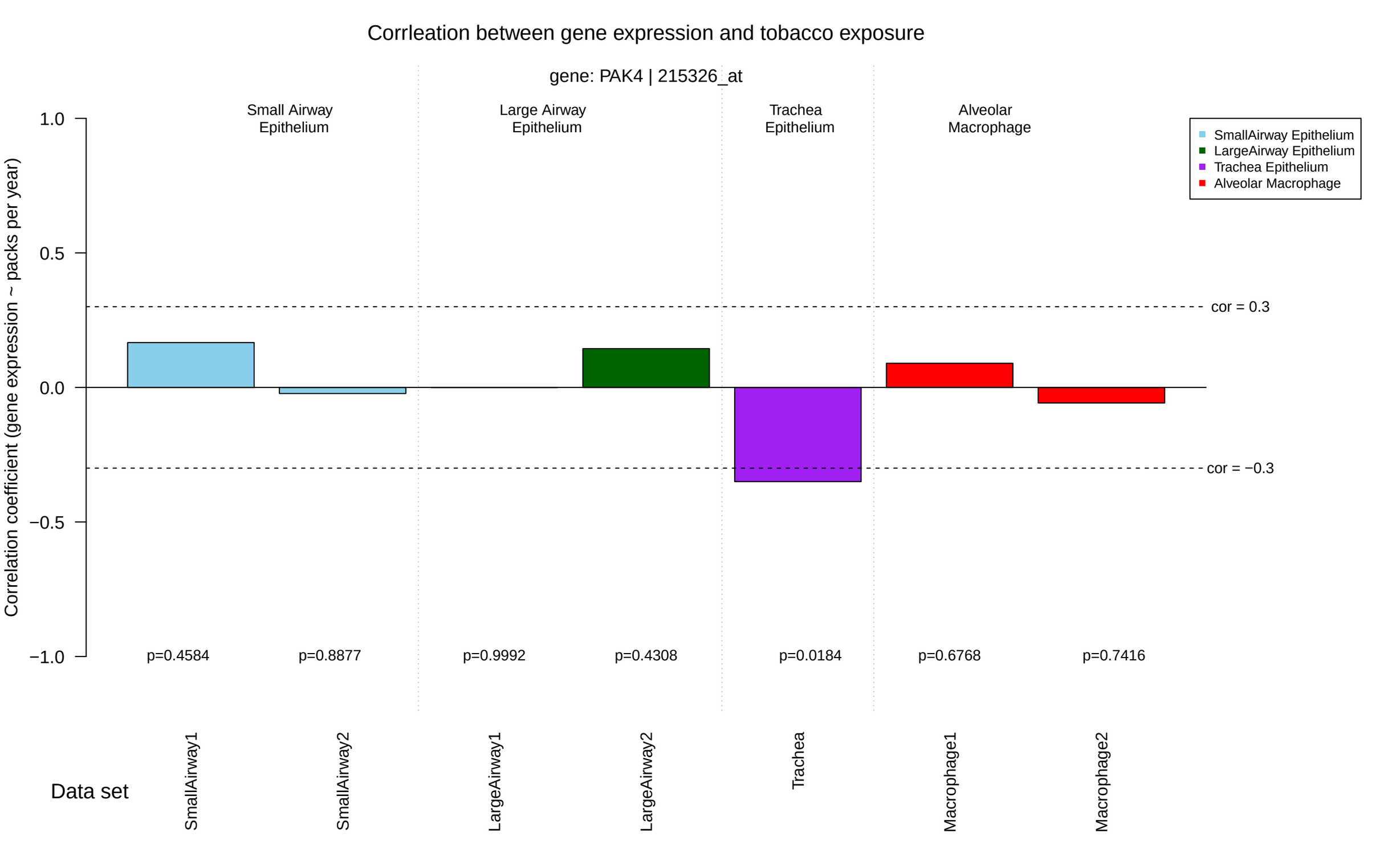

set: LargeAirway1 (p=0.4666)
PAK4 | 215326_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8537)
PAK4 | 215326_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2749)
PAK4 | 215326_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1345)
PAK4 | 215326_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9073)
PAK4 | 215326_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK4 | 215326_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4584
p=0.8877
p=0.9992
p=0.4308
p=0.0184
p=0.6768
p=0.7416
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2